

Wave motion or Travelling waves

Types of waves

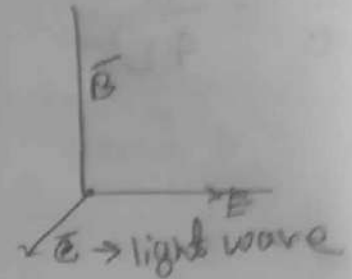
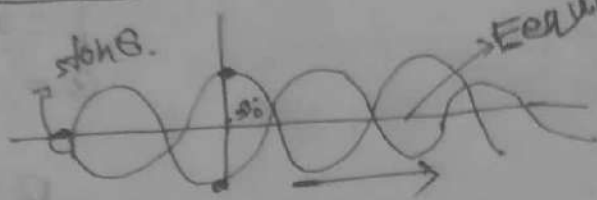
মেকানিক্যাল

ইলেক্ট্রোম্যাগনেটিক্যাল

মেকানিক্যাল:

(i) Transverse: (water wave, light wave)

A & B = 0



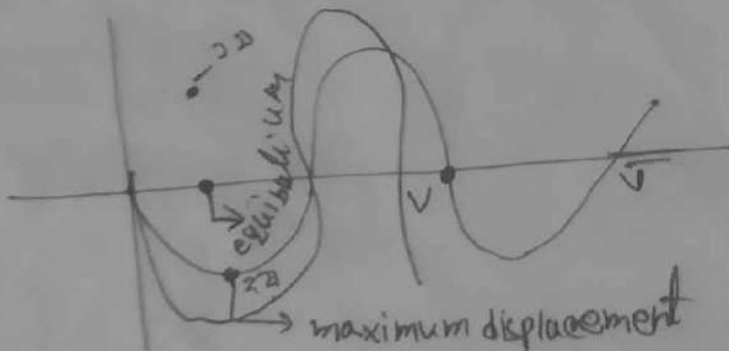
(ii) Longitudinal: (sound wave)

←→ motion of particle

→ velocity

[light wave medium ছাড়া propagate করতে পারে।]

water wave:



Simple Harmonic wave

২২ থেকে ২২ পর্যন্ত আসা মাধ্যমের সময় টি হল Time period.
এটি দুই কম্পনের দৈর্ঘ্য

λ = wave length
 v = wave velocity

λ > constant
medium এর উপর
to depend করে।

energy বৃদ্ধি amplitude
বৃদ্ধি করে।

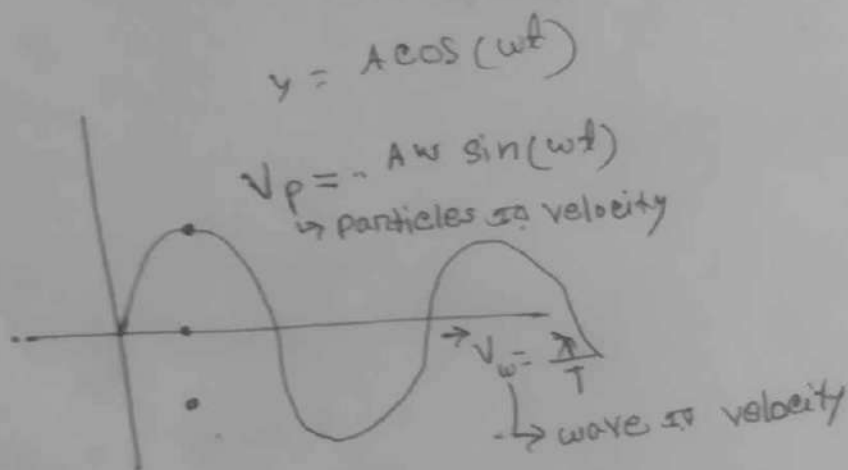
S.H.M & time period same হবে না।

$$v = \frac{\lambda}{T} = \text{constant.} \quad [\text{medium chng হল } v \text{ চকু হবে}]$$

→ wave velocity

maximum displacement & velocity 0.

particle এর velocity চকু হয়।

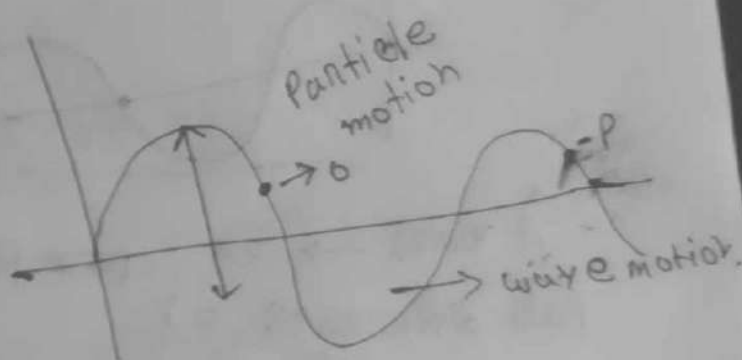


wave এর velocity
এবং particle এর
velocity এক হবে
না। কিন্তু medium
না থাকলে এক হবে
0 হবে।

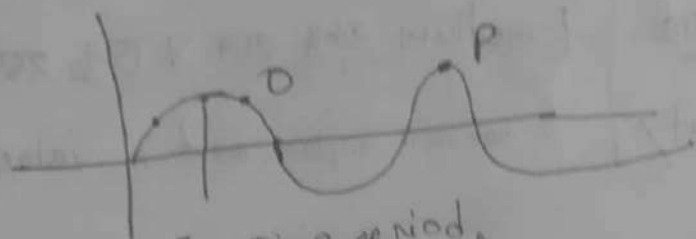
$$v = f \lambda$$

$$f = \frac{1}{T}$$

$$v = \frac{\lambda}{T}$$



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$T \rightarrow$ Time period
 $\lambda \rightarrow$ wave length
 $v \rightarrow$ wave velocity



λ सासडा $\rightarrow 2\pi$ (एक चक्र)
 path difference $= \lambda$
 phase difference $= 2\pi$

O and P, path difference x , phase difference $= \frac{2\pi x}{\lambda}$

therefore,

$$\phi = \frac{2\pi}{\lambda} \cdot x$$

$$y_2 = A \sin(\omega t - \phi)$$

$$= A \sin\left(\omega t - \frac{2\pi}{\lambda} x\right)$$

$$= A \sin\left(\frac{2\pi}{T} t - \frac{2\pi}{\lambda} x\right)$$

$$= A \sin 2\pi\left(\frac{1}{T} t - \frac{x}{\lambda}\right)$$

$$= A \sin 2\pi\left(\frac{v}{\lambda} t - \frac{x}{\lambda}\right)$$

$$\left[v = \frac{\lambda}{T} \text{ or } T = \frac{\lambda}{v}\right]$$

$$y = A \sin \frac{2\pi}{\lambda} (vt - x)$$

→ equation for travelling wave.

$$y = A \sin(\omega t)$$

$$v_p = \frac{dy}{dt} = A\omega \cos(\omega t)$$

$$v_p = 0 \Rightarrow \max.$$

$$y = A \sin\left(\frac{2\pi}{\lambda} (vt - x)\right)$$

$$\frac{dy}{dt} = A \left(\frac{2\pi}{\lambda} v\right) \cos \frac{2\pi}{\lambda} (vt - x)$$

$$\frac{dy}{dx} = -A \left(\frac{2\pi}{\lambda}\right) \sin \frac{2\pi}{\lambda} (vt - x)$$

→ slope of displacement curve / strain.

* disturbance ना থাকলে strain = 0.

$$\frac{dy}{dt} = - \frac{dy}{dx}$$

$$-\frac{dy}{dx} = -A \left(\frac{2\pi}{\lambda} \right) v \cos \frac{2\pi}{\lambda} (vt - x)$$

$$\frac{d^2y}{dx^2} = +A \left(\frac{2\pi}{\lambda} \right)^2 \sin \frac{2\pi}{\lambda} (vt - x)$$

$$\frac{d^2y}{dx^2} = -A \left(\frac{2\pi}{\lambda} \right)^2 \cos \frac{2\pi}{\lambda} (vt - x)$$

$$\frac{d^2y}{dx^2} = -v^2 \frac{d^2y}{dx^2} \rightarrow$$